

The Anatomy of the Dynamic Gap

Preview de la seccion; el resto del paper aparece solo como referencia

Contexto referenciado (no forma parte de la seccion)

Separacion. La seccion de separacion establece que en una poblacion homogenea de alta prevalencia el mejor diseno estatico obtiene Buq mientras que una estrategia dinamica con conteos obtiene al menos $u[1 - (1 - q)^{kG}]$ bajo el mismo objetivo.

Historias laminas. Los pools probados parten la poblacion testeada en atomos residuales disjuntos, uno por pool.

Lemma 1 (Los atomos residuales parten el soporte testeado). *En una familia laminar los atomos residuales son disjuntos dos a dos y su union es la union de los pools.*

Lemma 2 (Conteos de pool y de atomo se determinan mutuamente). *Los dos sistemas de restricciones recortan el mismo conjunto de mundos, y las sumas por subarbol invierten la resta que define los conteos residuales.*

Proposition 1 (Mejora de politica). *El rollout replanificado sobre una biblioteca laminar domina al greedy en todo estado alcanzable.*

1 The Anatomy of the Dynamic Gap

Section establishes that dynamic count-valued testing can strictly outperform static designs. This section asks a finer question: what is that advantage made of, and how much of it survives the structural restrictions that make the problem computationally tractable?

We decompose the gap along two orthogonal axes — the *observation channel* (binary versus count-valued) and the *design mode* (static versus adaptive) — and then measure separately the price of restricting attention to laminar pool histories. All quantities reported here are exact: they are obtained by enumerating the belief state, not by sampling.

Conventions. Throughout, $X_i = 1$ indicates that agent i is infected, $p_i = \Pr[X_i = 1]$, and $q_i = 1 - p_i$ is the prior probability that agent i is healthy. The regime of interest is $q_i < \frac{1}{2}$, in which healthy agents are the scarce resource. We write $r(t, X) = \sum_{i \in t} X_i$ for the count returned by an augmented test on pool t .

Two coarsenings of the count are natural, and they are not equivalent:

$$Y^{\text{inf}}(t) = \mathbf{1}\{r(t, X) > 0\}, \quad Y^{\text{heal}}(t) = \mathbf{1}\{r(t, X) < |t|\}.$$

The first is the classical group-testing channel and reports the presence of an infected agent; the second reports the presence of a *healthy* agent. In a welfare-maximizing setting the object of interest is the healthy population, and Y^{heal} is the binary channel we study. Every binary quantity below refers to Y^{heal} .

We also distinguish two certification rules, and the distinction turns out to be consequential rather than cosmetic.

Definition 1 (Certification rules). *Fix a testing history H with observed outcomes. Under credited-only certification, agent i contributes u_i to welfare if and only if i belongs to some tested pool t with observed count $r(t) = 0$. Under inference-based certification, agent i contributes u_i if and only if $X_i = 0$ in every assignment consistent with H .*

Credited-only certification is the hard-clearing convention: a deduction informs future decisions but does not itself generate reward. Inference-based certification counts an agent whenever the observed constraint system determines its status. Every agent credited under the first rule is certified under the second, but not conversely.

1.1 Two Orthogonal Ingredients

We first quantify the two ingredients separately. Consider $N = 10$ homogeneous agents with $q_i = 0.2$ and $u_i = 1$, no pool-size constraint, inference-based certification, and a testing budget B . Table 1 reports the exact optimum of each class, obtained by complete enumeration of the belief state.

B	static binary	dynamic binary	static count	laminar count	dynamic count
1	0.200	0.200	0.200	0.200	0.200
2	0.400	0.488	0.408	0.536	0.536
3	0.600	0.790	0.800	0.928	1.000

Table 1: Exact expected welfare by class, $N = 10$, $q_i = 0.2$, $u_i = 1$, inference-based certification. “Laminar count” restricts the dynamic count-valued optimum to histories in which any two pools are nested or disjoint.

Three observations follow. First, at $B = 1$ all five classes coincide, which fixes the base case. Second, at $B = 3$ the two ingredients are of comparable value in isolation: counting without adaptivity yields 0.800 and adaptivity without counting yields 0.790, an increase of 33% and 32% respectively over the static binary baseline. Neither dominates the other. Third, the two together yield 1.000, an increase of 67%, so the ingredients are complementary rather than redundant.

That counting alone should be worth as much as adaptivity alone is not obvious, and the next subsection explains where the static count-valued gain comes from.

1.2 Certification Rules and the Static Baseline

The value of the static count-valued class depends entirely on which certification rule is in force. Under credited-only certification, singleton designs are optimal in the high-prevalence regime.

Proposition 2 (Optimality of singleton designs under credited-only certification). *Let all agents be homogeneous with $q_i = q \leq \frac{1}{2}$ and $u_i = u$, and adopt credited-only certification. Then no static design of B tests achieves expected welfare greater than Buq , and B distinct singleton tests attain this value.*

Proof. Let $t_1, \dots, t_B$ be any static design. An agent is credited only if it lies in a pool whose observed count is zero, so by a union bound over pools the expected number of credited agents is at most $\sum_{j=1}^B |t_j| \Pr[r(t_j, X) = 0] = \sum_{j=1}^B |t_j| q^{|t_j|}$. For $g \geq 1$ the ratio of consecutive terms of $g \mapsto gq^g$ is $q(g+1)/g \leq 2q \leq 1$, so the sequence is nonincreasing and $\max_{g \geq 1} gq^g = q$. Hence the expected welfare is at most Buq . Singleton tests on distinct agents attain it, since each is credited independently with probability q . $\square$

Under inference-based certification the conclusion fails, and it fails for a structural reason: a count is an integer-valued linear measurement, and a small number of such measurements can identify more agents than they test individually.

Proposition 3 (Three count-valued tests identify four agents). *There is a static design of three count-valued tests on four agents that determines $X \in \{0, 1\}^4$ exactly, for every realization. Consequently, under inference-based certification with homogeneous parameters, the static count-valued optimum with $B = 3$ is at least $4uq$, which strictly exceeds the bound $3uq$ of Proposition 2.*

Proof. Label the agents a, b, c, d and take

$$t_1 = \{a, b, d\}, \quad t_2 = \{a, c, d\}, \quad t_3 = \{b, c, d\},$$

so that each agent belongs to a distinct subset of the tests. Writing r_1, r_2, r_3 for the observed counts,

$$r_1 + r_2 + r_3 = 2(X_a + X_b + X_c) + 3X_d,$$

whose parity equals the parity of X_d ; since $X_d \in \{0, 1\}$, this determines X_d . Given X_d , the three equations become $X_a + X_b = r_1 - X_d$, $X_a + X_c = r_2 - X_d$, $X_b + X_c = r_3 - X_d$, whose sum is $2(X_a + X_b + X_c)$ and therefore determines $X_a + X_b + X_c$; subtracting each pairwise sum recovers each variable individually. All four agents are thus determined, and the $4q$ of them that are healthy in expectation are certified. $\square$

The same statement is impossible for binary tests, by a counting argument that requires no design at all.

Proposition 4 (Binary channels cannot identify four agents with three tests). *No static design of three binary tests determines $X \in \{0, 1\}^4$ for every realization.*

Proof. Three binary tests induce at most $2^3 = 8$ distinct outcome vectors, while there are $2^4 = 16$ realizations. By pigeonhole two distinct realizations produce the same outcome vector; they differ in the status of some agent, which therefore cannot be determined. $\square$

Proposition 3 is an instance of the coin-weighing problem of Erdős and Rényi: B measurements that report subset sums identify on the order of $B \log B$ binary unknowns. Under inference-based certification the static count-valued optimum therefore grows superlinearly in the budget, rather than as Buq . Any separation argument that measures the dynamic advantage against Buq is correct under credited-only certification and compares against the wrong baseline under the inference-based rule. We adopt credited-only certification as the normative convention of the paper and record the alternative explicitly, since the choice determines the value of an entire class.

1.3 The Cost of Laminarity

Laminar histories admit exact inference (Section), and with count-valued tests the structure is even stronger: every tested pool ends with a known count, so the tested population decomposes into residual atoms with known counts and the untested territory retains its product prior.

Proposition 5 (Whole atoms carry no information). *Let H be a laminar history of count-valued tests with residual atoms $\{D_t\}$ and residual counts $\{c(D_t)\}$, and let $V(H)$ denote the untested territory. Let s be a new pool that is the union of a subfamily $\mathcal{A}$ of whole residual atoms with a subset $v \subseteq V(H)$. Then*

$$r(s, X) = \sum_{D \in \mathcal{A}} c(D) + r(v, X),$$

so the observation $r(s, X)$ is informationally equivalent to $r(v, X)$ alone.

Proof. By Lemma 2(a) the history determines $r(D, X) = c(D)$ for every residual atom D . The atoms are pairwise disjoint and disjoint from $V(H)$ by Lemma 1, so counts add over the blocks of s . The first summand is a constant given H , and subtracting a constant from an observation preserves the induced partition of the sample space. $\square$

Proposition 5 has an algorithmic consequence: when searching over laminar continuations it suffices to consider fresh pools and strict subsets of a single atom, since enlarging a pool by whole atoms changes nothing. This reduces the laminar state to a multiset of (size, count) pairs together with the size of the untested territory, and it is the reason the laminar optimum remains computable at budgets where the unrestricted optimum does not: in our implementation the laminar solver reaches $B = 8$ in seconds, while the unrestricted solver does not terminate at $B = 4$.

Tractability, however, is not free. Table 1 shows that at $B = 3$ the laminar optimum is 0.928 against an unrestricted optimum of 1.000, a relative loss of 7.2%, while at $B \leq 2$ the restriction is costless. The optimal unrestricted policy crosses a previously tested pool, and the mechanism is worth stating explicitly.

Example 1. *Consider $N = 10$ homogeneous agents with $q_i = 0.2$, $u_i = 1$, and $B = 3$ under inference-based certification. The optimal policy first tests a pool $\{a, b, c\}$. On the outcome $r = 2$, exactly one of the three agents is healthy and each has posterior health probability $\frac{1}{3}$. The optimal second test is $\{a, b, d\}$, where d is untested — a pool that neither contains, is contained in, nor is disjoint from $\{a, b, c\}$.*

The crossing test performs two functions simultaneously. If it returns a count of three, then a , b and d are all infected, so c is healthy and is certified without ever being tested, and the remaining budget is not spent on d . If it returns a count of one, then exactly one of $\{a, b\}$ is healthy and so is d , which is certified; the last test resolves the pair. A laminar policy must choose between probing the tested group and probing fresh territory; the crossing test uses the residual information of the group to extract value from a new agent with the same test.

The loss reported here is a single instance and is evidence, not a bound; adversarial search over the same family has located instances with a ratio of 0.9069.

1.4 Where Policies Leave the Laminar Class

Example 1 raises the question of how often crossing matters. We answer it by classifying every decision taken by a policy at every reachable state, weighted by the probability of reaching that state.

Definition 2 (Crossing). *A pool t crosses a pool s if $t \cap s \notin \{\emptyset, t, s\}$. An action is crossing with respect to a history H if it crosses some pool already executed in H . Disjoint pools, descendants, ancestors and repetitions are not crossing.*

For a policy π and a class κ of actions we write $W_\kappa^\pi = \sum_H \Pr^\pi[H] \mathbf{1}\{\pi(H) \in \kappa\}$ for the probability mass of decisions of that class. Table 2 reports W_{cross}^π over a sweep of 54 instances with $N \in \{4, 5, 6\}$, $B \in \{1, 2, 3\}$, pool-size constraint $G \in \{2, 3\}$ and $q \in \{0.05, 0.45, 0.90\}$, comprising 872 decisions in total.

Policy	crossing mass	mean local regret
myopic S_0	0.0026	0.0160
rollout	0.0215	0.0000
unrestricted optimum	0.0367	0.0026

Table 2: Probability mass of crossing decisions, and mean local regret against the rollout Q -function, averaged over 54 instances.

The crossing mass is strictly increasing in the quality of the policy: the optimum crosses roughly fourteen times as often as the myopic policy. The laminar restriction therefore binds precisely on the policies that are worth restricting, which is the empirical counterpart of Example 1.

The second column records a further observation. The rollout policy maximizes Q^g by construction and so has zero local regret; the unrestricted optimum does not, because Q^g evaluates an action under a greedy continuation rather than an optimal one. The magnitude of that regret quantifies the residual between rollout and optimality.

1.5 Planning without Solving

Exact dynamic optimization is infeasible beyond small instances, which motivates policies that plan without solving. The rollout policy scores each action by its value under a greedy continuation,

$$Q_b^g(H, t) = \mathbb{E}[r(H, t, R) + V_{b-1}^g(H') \mid H, t],$$

executes the maximizer, and re-plans at the next state rather than continuing greedily. Proposition 1 guarantees $V^\pi \geq V^g$ at every reachable state.

We verify the implementation rather than assume it. The hypothesis that the greedy action belongs to the rollout candidate set is checked state by state, as is the dominance conclusion. Expected values are computed by two independent procedures: a backward recursion over the belief tree, which branches on observed counts and weights by conditional probabilities, and a forward enumeration over all 2^N latent profiles, which runs the policy deterministically on each profile and averages under the prior. No probability appears inside the second computation. Over the test battery the two agree to within 2.22×10^{-16} .

On the instance $N = 5$, $q_i = 0.3$, $u_i = 1$, $B = 3$, $G = 2$, the myopic policy attains 0.900 and the rollout attains 1.011, an improvement of 12.3%. The myopic value coincides digit for digit with the static optimum, since a myopic policy on homogeneous parameters never conditions on what it observes and is therefore a fixed design in disguise.

A diagnostic for surrogate objectives. The same machinery supplies a falsification test for candidate scoring functions. In the family with homogeneous parameters and small q , the plan that covers the population with k disjoint pools of size G and then locates a healthy agent by binary search attains, under credited-only certification,

$$u \left(1 - (1 - q)^{kG} \right), \quad k = \max\{0, B - \lceil \log_2 G \rceil - 1\},$$

where the subtracted test is the one that converts a located agent into credited utility. At $(q, G, k, B) = (0.05, 16, 2, 7)$ this gives $0.8063u$ against a singleton baseline of $0.35u$, a ratio of 2.30. A scoring function that does not value this plan cannot recover the separation, whatever its form. The myopic score S_0 fails exactly two of the nine trajectory checks: it does not value opening fresh territory, since a pool of size g is scored by q^g , and it is blind to the remaining budget. These are precisely the two properties a planning score must supply.

1.6 Open Questions

1. Is the near-equality between the value of counting alone and the value of adaptivity alone a coincidence of these parameters, or does it reflect a structural relation between the two relaxations?
2. Under inference-based certification, what is the exact growth rate of the static count-valued optimum, and does the coin-weighting construction remain optimal when utilities are heterogeneous?
3. Proposition 2 is stated for homogeneous parameters. Does the singleton bound survive heterogeneous q_i and u_i , where a pool contributes $(\prod_{i \in t} q_i) \sum_{i \in t} u_i$?
4. Can the laminar class be relaxed to admit a bounded number of crossings while retaining exact inference, and does the resulting class recover the 7.2%?
5. Given that the unrestricted optimum does not maximize Q^g , is there a continuation policy, cheaper than optimal, whose induced Q -function is maximized by the optimum?
6. Does a budget-aware surrogate exist that passes all nine trajectory checks while remaining strictly cheaper to evaluate than the rollout oracle?

Taken together, these results suggest that the advantage of dynamic count-valued testing is not a single phenomenon but the composition of three separable ones: the resolution supplied by

integer-valued observations, the conditioning supplied by adaptivity, and the freedom to test pools that cut across the history. The first two are of comparable magnitude and combine superadditively; the third is small in mass but concentrated on exactly the decisions that distinguish good policies from mediocre ones.